

PolySkill Abstract Class

The Abstract Domain Class, defining a "schema" for shopping sites.

```
class AbstractShoppingSite:
    def search_product(self, query: str):
        """Searches for a product."""

    def add_to_cart(self, item_id: str, quantity: int):
        """Adds a specified item to the shopping cart."""

    def checkout(self):
        """Initiates the checkout process."""

    ## -----
    ## Compositional Skills
    ## -----

    def find_and_add_to_cart(self, query: str, item_id: str):
        """Searches for a product and adds it to the cart."""
        self.search_product(query)
        self.add_to_cart(item_id)

    def purchase_item(self, query: str, item_id: str):
        """Finds a specific product, adds it to the cart, and starts
checkout."""
        self.find_and_add_to_cart(query, item_id)
        self.checkout()
```

PolySkill Implementation

```
class AmazonWebsite(AbstractShoppingSite):
    def search_product(self, query: str):
        click(search_bar_id)
        fill(search_bar_id, query)
        keyboard_press('Enter')

    def add_to_cart(self, item_id: str, quantity: int):
        click(add_to_cart_button_id)

    def checkout(self):
        """NOTE: Implements checkout as a composite action: first
cart, then clicks checkout."""
        self.view_cart()
        click(checkout_button_id)
```



```
class TargetWebsite(AbstractShoppingSite):
    def search_product(self, query: str):
        fill(search_bar_id, query)
        click(search_button_id)

    def add_to_cart(self, item_id: str, quantity: int = 1):
        click(add_button_id)
        fill(quantity_bar_id, quantity)
        click(add_to_cart_button_id)

    def checkout(self):
        click(checkout_button_id)
```

